

INTRODUCTION Some affixes assign an accent or stress to roots that they attach to. Most known cases involve local accentuation: prefixes assign root-initial accents and suffixes assign root-final accents (Kurusu 2001; Revithiadou 2008). However, Japanese speakers started using a suffix [-zu] to form group names, and a recent elicitation study by Kawahara & Wolf (to appear) shows that this suffix assigns a root-initial accent, accompanied by lengthening of the root-final syllable (e.g. [usagi] ‘rabbit’ → [úsagii-zu] ‘a group of people who like rabbits’). This talk expands on their work and reports a wellformedness judgment experiment on this suffix [-zu]. This study improves on Kawahara & Wolf in several aspects: (i) Kawahara & Wolf used real words, whereas this study used nonce-words to test the real productivity of root-initial accentuation, (ii) this study controls for syllable and morphological structures and (iii) uses auditory stimuli rather than relying on orthography. Our result overall supports the root-initial accenting nature of [-zu].

STIMULI The experiment had four conditions, with ten items per each condition. The first two conditions were zu-words derived from nonce roots with four light syllables, both unaccented and accented (e.g. [rasobowa] → [rásobowaa-zu], [sokúdemu] → [sókudemuu-zu]). Both accented and unaccented roots were included because Kawahara & Wolf (to appear) showed that unaccented roots are more prone to show initial-accenting. The other two conditions were monomorphemic nonce words that had the same syllable structure as the zu-words, one with antepenultimate accent (e.g. [wabinoháada]), which is the default accent placement in Japanese (Kubozono 2009), and the other with initial accent (e.g. [kúyamenáada]). The prediction was that Japanese speakers would accept initial-accented zu-words, but not with monomorphemic words.

RECORDING AND TESTING A native speaker of Japanese pronounced all the stimuli in a frame sentence in a recording booth, digitized at 44K sampling rate. The target items were extracted from the frame sentences, and the average amplitude was adjusted to 65dB by Praat (Boersma and Weenink 2009). The judgment task was administered online. The participants were asked to judge the naturalness of accents of the stimuli in a 1-to-5 scale, from “very unnatural” to “very natural”. In Part I, the participants judged the naturalness of monomorphemic stimuli with initial and antepenultimate accents. In Part 2, the participants were first presented with root forms, followed by zu-words, and were asked to judge the naturalness of the accents of the zu-words. The participants were asked to take the online test in a quiet environment. Fifty-one native speakers of Japanese completed the study.

RESULTS AND DISCUSSION The result appears in Figure 1 which shows the average naturalness ratings of the four conditions: monomorphemic initial accents, monomorphemic antepenultimate accents, initial-accented zu-words derived from unaccented roots, and initial accented zu-words derived from accented roots. Statistical comparisons are listed in Table 1. We observe that given monomorphemic words, Japanese speakers prefer antepenultimate accents to initial accents. The initial accents received much higher rating in zu-words than in monomorphemic words. This difference indicates that speakers do accept initially-accented forms if they are derived by [-zu], supporting the nature of [-zu] as an initially accenting suffix. The initial accenting in [-zu] words was nevertheless rated lower than the default antepenultimate accent in monomorphemic words, probably because zu-words are new in the language. We do not observe a difference between zu-words derived from unaccented roots and zu-words derived from accented roots; it is possible that our participants failed to pay attention to the derivational relationship between the roots and corresponding zu-words, and judge the naturalness of zu-words alone.

To summarize the results, our wellformedness judgment supports the initial accenting behavior of [-zu] with nonce-words. This study, together with Kawahara & Wolf (to appear), shows that affix-controlled accentuation can be non-local, expanding our typology of morphologically controlled accentuation.

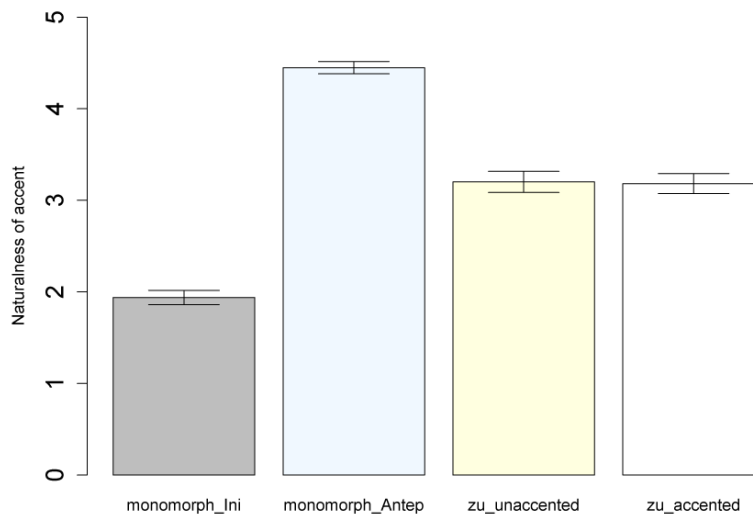


Figure 1: The averages of the naturalness ratings in four conditions: monomorphemic initial accents, monomorphemic antepenultimate accents, initial accented zu-words derived from unaccented roots, initial accented zu-words derived from accented roots. The error bars represent 95% confidence intervals.

Table 1: The results of mixed linear model analyses with the 95% confidence intervals of the coefficients. Contrast I=monomorphemic initial accents vs. zu-words; Contrast II=monomorphemic antepenultimate accents vs. zu-words; Contrast III=zu-words from accented roots vs. zu-words from unaccented roots. The 95% confidence intervals were calculated by the Markov chain Monte Carlo method.

	lower CI	upper CI	<i>t</i>	<i>p</i>
Contrast I				
monomorphemic initial vs. zu	1.0740	1.317	19.698	< .001
Contrast II				
monomorphemic antepenultimate vs. zu	-1.317	-1.084	-20.13	< .001
Contrast III				
unaccented vs. accented	-0.3046	0.0676	-1.221	<i>n.s.</i>